



Project Proposal

Topic: New Zealand Crime Rate Forecasting
Application

Subject: 158.888 Information Technology
Professional Project

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Introduction

Crime prevention and resource planning rely heavily not only on understanding of the location and time of occurrence of crime, but also on estimation of next occurrence. In New Zealand, the Police enable public accessibility to recorded crime data through their official website policedata.nz. This included the Recorded Crime Victims Statistics (RCVS) as well as Recorded Crime Offenders Statistics (RCOS), which are updated monthly and organized by offence category, Police District and time period (Crime Statistics Publications, n.d.).

Although the data is open and regularly refreshed, it is not widely available in a form that provides forward-looking forecasts at a regional level. To address this gap, this project proposes the development of a Crime Rate Forecasting Application for New Zealand. The application will ingest historical crime data from Police Data NZ, model crime trends by region and offence type, and generate short-term forecasts at both the regional and national levels.

Consequently, the resulting application will present these forecasts through an accessible dashboard, making it easier for users to explore historical patterns, compare regions and interpret likely future trends in crime across New Zealand.

Problem Statement

Crime is a primary social issue worldwide. Furthermore, regional crime trends in New Zealand vary considerably in both direction and magnitude. Understanding these changes over time can support evidence-based planning, resource allocation, and policy development for researchers, communities, government agencies, and the wider public.

Therefore, this project seeks to address this gap through the development of a lightweight, reproducible, and transparent forecasting application that visualises and predicts crime trends at both regional and national levels. The functions of the application include consolidating monthly recorded crime data into clean and analysis-ready dataset, model historical trend, seasonality and regional variation using machine-learning means and produce interpretable short-term forecasts at both regional and national level via simple and interactive interface.

Project Objective

The aim of the project is to design and implement a data-driven application that forecasts crime trends in New Zealand at both the regional and national level, using publicly available Police Data NZ statistics.

To achieve this aim, the following objectives have been established.

- Acquire, clean, and structure historical crime data from Police Data NZ into a consistent time-series format by region and offence category.
- Conduct exploratory data analysis to identify trend, seasonality, and regional differences in recorded crime.
- Build forecasting models to predict future crime rates and volumes.
- Develop and compare baseline and advanced forecasting models for regional and national crime counts.
- Evaluate model performance using standard forecasting metrics (MAE, RMSE, MAPE) via time-series cross-validation.
- Build a Python-based application that visualises historical trends and forecasted crime rates by region and nationally.

Data Source

To support the forecasting objectives outlined above, reliable and publicly available crime datasets will be utilized. The primary data source will be New Zealand Police's public data portal, policedata.nz, which provides reports and downloadable data. Key datasets of interest include:

- RCVS — victimisations by offence type, location, and time period.
- RCOS — offences that do not require an identified victim.
- Supplementary datasets from data.govt.nz for historical records.

Data will be aggregated at a monthly frequency and analysed at the Police District level, which represents New Zealand's standard policing boundaries. This level of aggregation provides an appropriate balance between geographical granularity and statistical reliability. Where relevant, Stats NZ population estimates may be incorporated to calculate crime rates per capita in addition to raw counts.

Methodology

Following data acquisition, the project will progress through the key stages of data collection, data handling, exploratory data analysis, forecasting and application development.

1. Data Collection

- Download monthly crime statistics from resources listed in data source section.

2. Data Handling

- Select relevant fields needed and handle missing values, duplications, inconsistent regional naming and time periods.
- Transform the dataset into a structured time-series format suitable for forecasting analysis.

3. Exploratory Data Analysis

- Identify trend and seasonal patterns per region and offence type.
- Comparative visualisation of regional crime patterns and national aggregates.

4. Forecasting Models

- Machine-learning models with lag features will be used for time-series forecasting.
- Generate forecasts for both regional crime trends and national crime volumes.

5. Application Development

- A Python-based application to visualise historical trends and forecasts.
- Region selector and offence-type filters, with national roll-up view.

Tools and Technology

Python will be the main programming language used for this project. Supported libraries such as Pandas and NumPy will be utilized for data handling. The project will use Plotly for visualization. Forecasting models implemented through Statsmodels, Prophet, and scikit-learn will be evaluated and compared to determine the most effective approach for predicting crime trends. Streamlit will be applied for application interface. Development will be conducted in Visual Studio Code (VS Code), with GitHub used for version control and project management.

Project Scope

The project focuses exclusively on New Zealand. With historical data from the Police covering recorded crime by region and offence category, the forecast of time-series data will be focusing on the Police District and national level as boundaries.

In addition, the application will generate short-term forecasts with a prediction horizon of up to three months. Furthermore, a functional dashboard will be developed to facilitate exploratory analysis and improve the interpretability of forecast results.

Project Timeline

Phase	Week	Deliverables
Data Collection	Week 1 – 2	Download raw data and store them locally.
Data Handling	Week 3 – 4	Clean data, handle missing or inconsistent value, aggregate data and so on.
Feature Engineering	Week 5 – 7	Conduct EDA analysis on region and seasonal pattern as well as create lagged and rolling features.
Forecast Model	Week 8 – 9	Compare and contrast different models from Statsmodel, scikit-learn and Prophet using standard metrics.
Application Interface	Week 10 – 11	Develop an interactive dashboard highlighting regional and national crime forecast in New Zealand.
Testing, documentation and video presentation	Week 12 – 14	Test and deploy the application before final report and video presentation.

Expected Outcome

Upon completion, the project is expected to deliver a cleaned and reusable crime dataset covering New Zealand regions and offence categories. In addition, an interactive dashboard will enable users to visualise historical crime patterns and short-term forecasts at both regional and national levels. Furthermore, the application will support comparative analysis across regions, providing greater insight into spatial crime trends and their potential future development.

Risk and Limitation

While forecasting results may provide valuable insights for policymakers and planners, they should be interpreted as statistical projections of historical trends rather than precise predictions of future criminal activity. This is because recorded data may reflect reported or recorded offences but not the true incidence of crime happening.

Risk	Impact	Mitigation
Data aggregation level is too coarse for meaningful regional forecasting.	High	Use Police District-level data and evaluate details of data during exploratory analysis.
Low crime counts in small regions create unstable forecast.	Medium	Apply smoothing techniques, aggregate categories where appropriate, and compare model robustness across regions.
Project timeline exceeds available project duration.	Medium	Maintain a clearly defined project scope focused on monthly regional and national forecasting.

References

Crime Statistics Publications. (n.d.). New Zealand Police. <https://www.police.govt.nz/about-us/publications-statistics/data-and-statistics/crime-statistics-publications>